SECSR2043 OPERATING SYSTEMS
[20 Marks]

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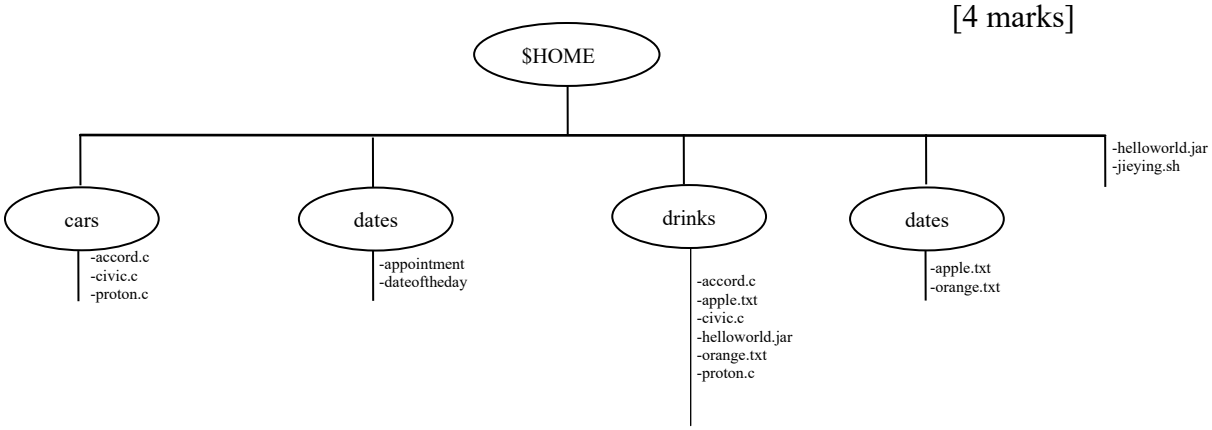
Instruction: Please answer all of the following questions. Whenever the ✋ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **$HOME** directory.

[4 marks]



Print screen the script that you type;

```
#!/bin/bash

echo "Hello World" > helloworld.jar

mkdir cars
mkdir dates
mkdir fruits
mkdir drinks

cd cars
echo "Honda Accord" > accord.c
cp accord.c civic.c
echo proton > proton.c

cd../dates
date > dateoftheday
cat dateoftheday > appointment

cd ../fruits
echo apple > apple.txt
cat apple.txt > orange.txt

cd ../drinks
cp ../cars/*.* .
cp ../fruits/*.* .
cp ../*.jar .
```

Then draw the tree

```
jieying@secr2043:~$ tree
.
├── bye.txt
├── cars
│   ├── accord.c
│   ├── appointment
│   ├── civic.c
│   ├── dateoftheday
│   └── proton.c
├── dates
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── echo
├── echo.c
├── fork_wait
├── fork_wait.c
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── greeting.txt
├── hello
├── hello.c
├── hello.txt
├── hello_bak.txt
├── helloworld.jar
├── jieying.sh
├── mainprocess
├── mainprocess.c
├── nano.1444.save
├── os_lab
│   └── hello.txt
├── subprocess1
├── subprocess1.c
├── subprocess2
├── subprocess2.c
├── sum
└── sum.c
```

b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
#!/bin/bash

read -p "Enter the file extension (without the dot): " ext

echo "Files with .${ext} extension:"
find . -type f -name "*.${ext}"

count=$(find . -type f -name "*.${ext}" | wc -l)
echo "Number of files with .${ext} extension: $count"
```

```
jieying@secr2043:~$ chmod +x ifiles.sh
jieying@secr2043:~$ ./ifiles.sh
Enter the file extension (without the dot): c
Files with .c extension:
./echo.c
./fork_wait.c
./sum.c
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./hello.c
./subprocess1.c
./subprocess2.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
./mainprocess.c
Number of files with .c extension: 13
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

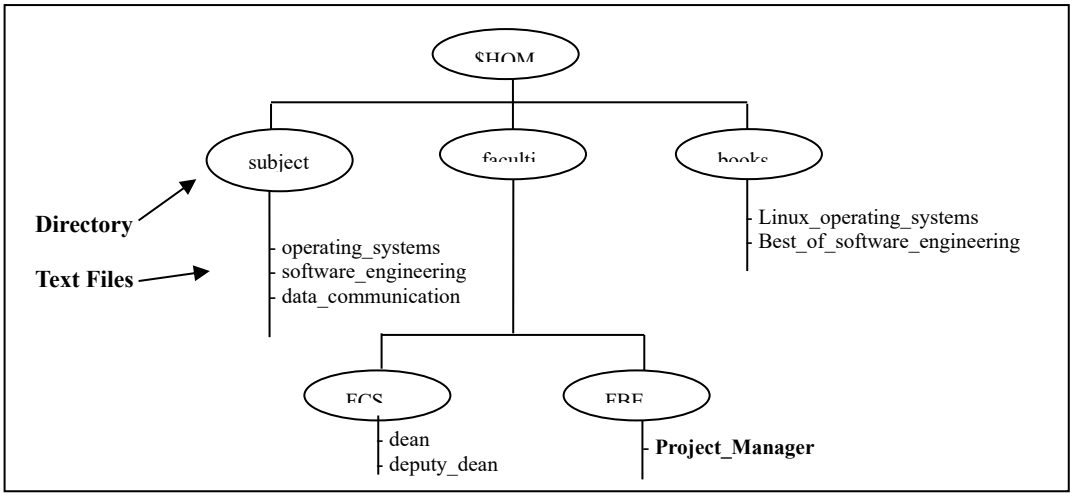


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```
#!/bin/bash

mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
```

```
jieying@secr2043:~$ tree faculties
faculties
├── FBE
│   └── Project_Manager
└── FCS
    ├── dean
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`. [2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
#!/bin/bash

chmod 744 $HOME/subjects

chmod 520 $HOME/books/Best_of_software_engineering

chmod 607 $HOME/faculties/FCS

chmod 031 $HOME/faculties/FBE/Project_Manager
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr—r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

**** All the Best for Final Exam ****